

A non-Baire translation-invariant ideal with a perfect translation-almost-disjoint family

An affirmative construction for a question of Filipów and Tryba

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Abstract

We give a ZFC construction intended to answer Question 1 of Filipów and Tryba, *Densities for sets of natural numbers vanishing on a given family*, affirmatively. For every free ultrafilter $\mathcal{U}$ on ω , an explicit partition into finite intervals defines a proper translation-invariant ideal $\mathcal{I}_{\mathcal{U}}$ without the Baire property. There is a continuous injection $x \mapsto A_x$ from 2^ω into $\mathcal{I}_{\mathcal{U}}^+$ such that $A_x \cap (A_y + k)$ is finite whenever $x \neq y$ and $k \in \mathbb{Z}$. Thus the positive family can be perfect, and its translated intersections can be genuinely finite rather than merely ideal-small. The construction combines ultrafilter block supports with widely separated labels on a binary tree. A direct fusion argument proves the category assertion without relying on a preservation claim about arbitrary continuous preimages. We also establish quantitative intersection bounds, maximal ultrafilter traces, a rich maxitive abstract upper density with this exact null ideal, and an extension to countable permutation actions. Exact finite tests accompany the proof; they check its explicit arithmetic and fusion invariants, not the infinitary assertions. The literature search found the question explicitly posed in the original source, but does not constitute a definitive priority determination.

Keywords. Ideals on the integers; Baire property; free ultrafilters; translation almost disjointness; abstract upper densities.

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1 The problem, the result, and its status

1.1 Random selection and problem selection

The mathematical area was chosen before the problem. A single call to a pseudorandom generator, seeded with 128 bits from the operating system, selected index 22 from a fixed, zero-indexed list of 24 mathematical areas. The result was *set theory*. The seed, the complete list, and reproduction instructions appear in Appendix A and in `random_selection.json`. There was no reroll of the area.

Within set theory, ideals on a countable set offered a useful balance: the motivating question is genuinely infinitary, but potential witnesses can be built from finite blocks, binary trees, and ultrafilters. The selected problem is Question 1 of Filipów and Tryba [1]. In the notation developed below, its content is the following.

Problem 1.1 (Filipów–Tryba, Question 1). Find a translation-invariant proper ideal $\mathcal{I}$ on ω for which $\mathcal{I}$ lacks the Baire property and $\mathcal{I}^+$ contains a translation-almost-disjoint family of size $\mathfrak{c} = 2^{\aleph_0}$.

The original uses the positive integers. Passing between ω and the positive integers by $n \mapsto n + 1$ does not change this question. The translation conventions and the harmless finite boundary effects are made explicit in Section 2.

1.2 Main statement

Theorem 1.2 (Main construction). *In ZFC, for every free ultrafilter $\mathcal{U}$ on ω there exist a proper ideal $\mathcal{I}_{\mathcal{U}} \subseteq \mathcal{P}(\omega)$ and a continuous injection*

$$\Phi : 2^\omega \longrightarrow 2^\omega, \quad \Phi(x) = A_x,$$

with the following properties:

- (i) $\mathcal{I}_{\mathcal{U}}$ contains Fin, is translation invariant, and does not have the Baire property.
- (ii) Every A_x is $\mathcal{I}_{\mathcal{U}}$ -positive.
- (iii) If $x \neq y$ and $k \in \mathbb{Z}$, then $A_x \cap (A_y + k)$ is finite. If $k \neq 0$, this conclusion also holds for $x = y$.
- (iv) $\mathcal{I}_{\mathcal{U}}$ is tall. More strongly, its restriction to any positive set does not have the Baire property in the corresponding Cantor space.

The image of Φ is a compact perfect family of cardinality $\mathfrak{c}$. Both the block partition and the map Φ are independent of $\mathcal{U}$.

The proof is given in Sections 3–5. No additional hypothesis such as the continuum hypothesis is used. The construction is explicit *relative to a free ultrafilter*; this is not a claim that membership in the ideal is algorithmically decidable.

The theorem answers the existential question by strengthening its witness. An ideal-small intersection is allowed in the problem, whereas our intersections are finite. A family of continuum cardinality is requested, whereas our family is the continuous image of Cantor space. These stronger properties will also make it possible to prescribe positive values of an associated abstract density.

1.3 What is, and is not, being claimed

The original article was published in 2020; its arXiv version was posted in 2023. Question 1 is visible on page 7 of the ten-page preprint [1]. A targeted search on September 20, 2026 checked this source, the author's publication and citation lists, and related accessible later papers, including [5, 6]. No resolution of this particular question was located. These checks do not exclude an unindexed paper, an unpublished argument, or another treatment under different terminology. A search record is included in the archive.

The mathematical contribution claimed here is the complete written construction and proof, not a verified assertion of priority. The note has not been refereed or checked in a proof assistant. The explicit formulas and several finite versions of the combinatorial invariants were checked by an executable program. This provides protection against indexing errors but is not a substitute for the proofs.

The category criterion used below is classical, and the maximal-family method for constructing abstract densities is established in the literature [3, 2, 1]. These ingredients are not presented as new. The later questions in [1] about all invariant ideals or translation-maximal ideals are not answered by this construction.

2 Preliminaries and translation conventions

Write $\omega = \{0, 1, 2, \dots\}$ and identify a subset of a countable set X with its characteristic function in 2^X . This endows $\mathcal{P}(X)$ with the product topology. After any enumeration of X , this is Cantor space. A subset of this space has the *Baire property* if its symmetric difference with an open set is meager. A meager set is a countable union of nowhere dense sets.

Definition 2.1. An *ideal* on X is a family $\mathcal{I} \subseteq \mathcal{P}(X)$ that contains all finite subsets of X , is hereditary, is closed under finite unions, and does not contain X . Its positive sets are $\mathcal{I}^+ = \mathcal{P}(X) \setminus \mathcal{I}$. It is *tall* if every infinite $D \subseteq X$ has an infinite subset belonging to $\mathcal{I}$.

Every ideal is invariant under finite symmetric differences. Indeed, if $A \in \mathcal{I}$ and F is finite, then $A \triangle F \subseteq A \cup F \in \mathcal{I}$; the converse follows by applying the same observation again. We write $A =^* B$ when $A \triangle B$ is finite and $A \subseteq^* B$ when $A \setminus B$ is finite.

For $k \in \mathbb{Z}$ define the truncated shift

$$\tau_k(A) = (A + k) \cap \omega. \quad (1)$$

It preserves unions and inclusions. For all integers h, k ,

$$\tau_k(\tau_h(A)) =^* \tau_{k+h}(A). \quad (2)$$

To check this, the only possible loss is caused by an element shifted below zero at the intermediate step and shifted back into ω at the next step. Such original elements lie in a fixed finite initial segment, depending on h and k . An ideal is *translation invariant* if $A \in \mathcal{I}$ is equivalent to $\tau_k(A) \in \mathcal{I}$ for every k .

Definition 2.2. A family $\mathcal{A} \subseteq \mathcal{I}^+$ is *$\mathcal{I}$ -translation-almost-disjoint*, or *$\mathcal{I}$ -TAD*, if

$$A \cap \tau_k(B) \in \mathcal{I} \quad (A, B \in \mathcal{A}, A \neq B, k \in \mathbb{Z}).$$

We call the stronger property that all these intersections are finite *finite translation almost disjointness*.

A free ultrafilter $\mathcal{U}$ on ω is a maximal proper filter containing all cofinite sets and no finite sets. In particular, for every $S \subseteq \omega$ exactly one of S and $\omega \setminus S$ belongs to $\mathcal{U}$. Its dual ideal is

$$\mathcal{U}^d = \{S \subseteq \omega : S \notin \mathcal{U}\}.$$

For example, if $S, T \notin \mathcal{U}$, their complements belong to $\mathcal{U}$, so $\omega \setminus (S \cup T) \in \mathcal{U}$ and $S \cup T \notin \mathcal{U}$. Membership in $\mathcal{U}$ is unaffected by finite changes.

Free ultrafilters exist in ZFC by extending the cofinite filter to an ultrafilter. No special ultrafilter, such as a selective ultrafilter or a P -point, will be needed. In all that follows, $\mathcal{U}$ is an arbitrary fixed free ultrafilter.

3 An elementary category criterion for ideals

The next criterion is the form of the classical category characterization of ideals needed here; compare Talagrand [3] and its formulation in [1, Theorem 1.7]. We include the proof to make the category step inspectable.

Lemma 3.1. *If a proper ideal on a countably infinite set has the Baire property, then it is meager.*

Proof. Enumerate the ground set as ω . Suppose that $\mathcal{I}$ has the Baire property and is not meager. There is a nonempty basic cylinder $[s]$ in which $\mathcal{I}$ is comeager. For every binary word t of length $|s|$, flipping a suitable subset of the first $|s|$ coordinates gives a homeomorphism from $[s]$ onto $[t]$. It preserves $\mathcal{I}$ because $\mathcal{I}$ is invariant under finite changes. Thus $\mathcal{I}$ is comeager in every such cylinder, and hence in all of 2^ω .

The complementation map $A \mapsto \omega \setminus A$ is a homeomorphism. Its image of $\mathcal{I}$ is therefore also comeager. This image is disjoint from $\mathcal{I}$: otherwise an ideal would contain both a set and its complement, and so would contain ω . Two comeager subsets of Cantor space cannot be disjoint, by the Baire category theorem. This is a contradiction. $\square$

Lemma 3.2 (Finite-block category criterion). *A proper ideal $\mathcal{I}$ on ω has the Baire property if and only if there are integers*

$$0 = t_0 < t_1 < t_2 < \cdots$$

such that no member of $\mathcal{I}$ contains infinitely many whole intervals $[t_n, t_{n+1})$.

Proof. Suppose first that $\mathcal{I}$ has the Baire property. By Lemma 3.1, it is contained in $\bigcup_j K_j$, where each $K_j \subseteq 2^\omega$ is closed and nowhere dense.

We recursively choose $t_{n+1} > t_n$ and a binary word u_n on the coordinates $[t_n, t_{n+1})$ so that

$$[v u_n] \cap (K_0 \cup \cdots \cup K_n) = \emptyset \quad \text{for every } v \in 2^{t_n}. \quad (3)$$

Here $v u_n$ denotes concatenation. There are only finitely many such prefixes v . Start with an empty proposed common suffix. For the first prefix, extend the suffix until the resulting cylinder avoids the finite union of the K_j 's. For the next prefix, extend the same suffix further to obtain avoidance for that prefix as well. Previously obtained avoidance persists under extension. After processing every prefix, add a coordinate if necessary to make the suffix nonempty. Now set t_{n+1} equal to the resulting total length. Nowhere density justifies every extension.

If a point $z \in 2^\omega$ agrees with u_n on infinitely many of these intervals, it belongs to none of the K_j . For any fixed j , choose a matching $n \geq j$ and apply (3) to the actual prefix of z .

Suppose, toward a contradiction, that some $A \in \mathcal{I}$ contains infinitely many full intervals $[t_n, t_{n+1})$. On those intervals choose a subset $D \subseteq A$ whose characteristic function is u_n , and put no points into D elsewhere. Then D avoids every K_j , so $D \notin \mathcal{I}$. This contradicts heredity.

Conversely, suppose the stated partition exists. For each m , the set

$$E_m = \{A \subseteq \omega : [t_n, t_{n+1}) \not\subseteq A \text{ for every } n \geq m\}$$

is closed and nowhere dense. It is closed because each single-block condition is clopen. It has empty interior because any basic cylinder can be extended by putting an entire sufficiently late block into the set. The hypothesis gives $\mathcal{I} \subseteq \bigcup_m E_m$. Thus $\mathcal{I}$ is meager, and in particular has the Baire property. $\square$

The same statement holds on any countably infinite set after choosing an enumeration. To refute the Baire property, it therefore suffices to show that every finite-block partition has an infinite union of blocks belonging to the ideal. Our next lemma proves a somewhat stronger assertion.

4 Block-support ideals and the fusion argument

4.1 Block support and countably many transformations

Let X be a countably infinite set and let $(B_n)_{n \in \omega}$ be a partition of X into nonempty finite blocks. Define

$$\sigma(A) = \{n : A \cap B_n \neq \emptyset\}. \quad (4)$$

The simple block-support ideal is

$$\mathcal{J}_{\mathcal{U}} = \{A \subseteq X : \sigma(A) \notin \mathcal{U}\}.$$

Because $\sigma(A \cup D) = \sigma(A) \cup \sigma(D)$, this is an ideal. Notice that being small here does not mean having few points in each block. It means avoiding all points of a $\mathcal{U}$ -large collection of blocks.

For a slightly more general construction, let $f_i : D_i \rightarrow X$, $i \in \omega$, be partial maps with finite fibers, and assume the identity map on X occurs in the list. Set

$$T_i(A) = f_i[A \cap D_i]$$

and define

$$\mathcal{I} = \{A \subseteq X : \sigma(T_i(A)) \notin \mathcal{U} \text{ for every } i \in \omega\}. \quad (5)$$

This is again an ideal: each T_i preserves inclusions and finite unions, sends finite sets to finite sets, and the identity term ensures $X \notin \mathcal{I}$. The finite-fiber assumption will be needed in the fusion argument, not in this elementary ideal verification.

Lemma 4.1 (Separated-block fusion). *For the ideal (5), let $(Q_j)_{j \in \omega}$ be any sequence of pairwise disjoint, nonempty finite subsets of X . There exists an infinite $H \subseteq \omega$ for which*

$$\bigcup_{j \in H} Q_j \in \mathcal{I}.$$

The Q_j need not cover X .

Proof. We first choose increasing indices $j_0 < j_1 < \dots$ with the following finite-stage invariant:

$$\sigma(T_i(Q_{j_n})) \cap \sigma(T_\ell(Q_{j_r})) = \emptyset \quad \text{if } r < n \text{ and } i, \ell \leq n. \quad (6)$$

Suppose $j_0, \dots, j_{n-1}$ have been selected. The set of previously used block indices, with transformations through stage n , is

$$H_n = \bigcup_{r < n} \bigcup_{\ell \leq n} \sigma(T_\ell(Q_{j_r})).$$

It is finite. Consequently $C_n = \bigcup_{b \in H_n} B_b$ is finite, and so is

$$F_n = \bigcup_{i \leq n} f_i^{-1}(C_n).$$

Preimages here are taken in the domains D_i . Finiteness follows from the finite-fiber hypothesis. Only finitely many of the disjoint sets Q_j meet F_n . Choose $j_n > j_{n-1}$ outside those exceptions. Then $T_i(Q_{j_n}) \cap C_n = \emptyset$ for every $i \leq n$, which is exactly what is needed for (6).

Separate the selected sets into two unions:

$$C_0 = \bigcup_{n \text{ even}} Q_{j_n}, \quad C_1 = \bigcup_{n \text{ odd}} Q_{j_n}.$$

For fixed i, ℓ , the intersection

$$\sigma(T_i(C_0)) \cap \sigma(T_\ell(C_1)) \tag{7}$$

is finite. Indeed, if a block index lies in this intersection, it is witnessed by an even selected stage n and an odd selected stage r . These stages are distinct. If $\max(n, r) \geq \max(i, \ell)$, condition (6), with the larger stage taken last, forbids the intersection. All remaining witnesses come from finitely many early selected sets, and their transformed block supports are finite.

If neither C_0 nor C_1 belonged to $\mathcal{I}$, there would exist indices i and ℓ with

$$\sigma(T_i(C_0)) \in \mathcal{U}, \quad \sigma(T_\ell(C_1)) \in \mathcal{U}.$$

Their intersection would belong to $\mathcal{U}$, contradicting its finiteness in (7) and the freeness of $\mathcal{U}$. Hence at least one of C_0, C_1 belongs to $\mathcal{I}$. Its defining parity supplies the required infinite index set H . $\square$

Corollary 4.2. *The ideal (5) does not have the Baire property and is tall. For every $D \in \mathcal{I}^+$, its restriction*

$$\mathcal{I} \upharpoonright D = \{A \subseteq D : A \in \mathcal{I}\}$$

does not have the Baire property as a subset of 2^D .

Proof. Apply Lemma 4.1 to each finite-block partition supplied by a putative application of Lemma 3.2. The resulting infinite union contradicts that criterion, so $\mathcal{I}$ cannot have the Baire property.

For tallness, enumerate an arbitrary infinite D by distinct points d_j and apply the fusion lemma to $Q_j = \{d_j\}$. It produces an infinite subset of D belonging to $\mathcal{I}$.

Finally, if D is positive, $\mathcal{I} \upharpoonright D$ is a proper ideal on D . Any finite-block partition of D is an eligible sequence (Q_j) in the fusion lemma, even though it does not cover X . The same contradiction to the category criterion in 2^D proves the last assertion. $\square$

4.2 Taking the translation core

Return to $X = \omega$. Use the partial maps $f_k(n) = n + k$ on the domains where $n + k \geq 0$, enumerated over $k \in \mathbb{Z}$. They are injective, and their induced transformations are the truncated shifts. Define

$$\mathcal{I}_{\mathcal{U}} = \{A \subseteq \omega : \sigma(\tau_k(A)) \notin \mathcal{U} \text{ for every } k \in \mathbb{Z}\}. \tag{8}$$

This is the translation core of $\mathcal{I}_{\mathcal{U}}$.

Proposition 4.3. *For every partition of ω into nonempty finite blocks, $\mathcal{I}_{\mathcal{U}}$ is a proper, tall, translation-invariant ideal without the Baire property. Its restriction to each positive set also lacks the Baire property.*

Proof. All assertions except translation invariance follow from Corollary 4.2. For invariance, suppose $A \in \mathcal{I}_{\mathcal{U}}$ and fix $h \in \mathbb{Z}$. For each k ,

$$\sigma(\tau_k(\tau_h(A))) =^* \sigma(\tau_{k+h}(A)).$$

The right side does not belong to $\mathcal{U}$, by (8), and finite changes do not affect ultrafilter membership. Thus $\tau_h(A) \in \mathcal{I}_{\mathcal{U}}$. Conversely, if $\tau_h(A) \in \mathcal{I}_{\mathcal{U}}$, apply the already established closure to $-h$. Equation (2) and invariance of ideals under finite changes recover $A \in \mathcal{I}_{\mathcal{U}}$. $\square$

Remark 4.4 (Why the category proof needs care). An embedding of a free ultrafilter into a trace of an ideal does not, by itself, prove that the whole ideal lacks the Baire property. The Baire property is not preserved by preimages under arbitrary continuous maps. For instance, every subset of a closed nowhere dense copy of Cantor space is meager in the ambient Cantor space, even when its pullback under the embedding has no Baire property. The fusion argument avoids this pitfall. It also uses a single parity choice for *all* transformations, not a potentially different parity for each shift.

The preservation of non-Baire behavior under translation cores is already part of the surrounding theory: see [1, Proposition 3.8] and the countable-intersection theorem attributed there to Plewik [4]. The point of the direct argument is to verify the specific ideal used here without leaving the preservation step implicit.

5 An explicit separated binary tree

5.1 Blocks and labels

We now choose the block partition that supplies continuum many positive sets. For $n \geq 0$ put

$$\begin{aligned} d_n &= 2^n, & L_n &= 4^n + 2^{n+1}, \\ b_n &= \sum_{j < n} L_j = \frac{4^n - 1}{3} + 2(2^n - 1), \\ B_n &= [b_n, b_{n+1}) \cap \omega. \end{aligned} \tag{9}$$

Thus $b_0 = 0$, the intervals partition ω , and each has length L_n . Let $2^{<\omega}$ denote the finite binary words. If s has length n , let $\text{val}(s)$ be its usual binary value between 0 and $2^n - 1$, with $\text{val}(\emptyset) = 0$. Assign the integer label

$$p_s = b_n + 2^n(1 + \text{val}(s)). \tag{10}$$

The labels at level n form the finite set

$$T_n = \{p_s : |s| = n\}.$$

They lie inside B_n , and

$$\min T_n = b_n + 2^n, \quad \max T_n = b_n + 4^n, \quad b_{n+1} - \max T_n = 2^{n+1}. \tag{11}$$

Consecutive labels at level n are 2^n apart. The gap between the last label of level n and the first label of level $n + 1$ is

$$\min T_{n+1} - \max T_n = 2^{n+2}. \tag{12}$$

n	b_n	b_{n+1}	$ T_n $	Labels in T_n
0	0	3	1	1
1	3	11	2	5, 7
2	11	35	4	15, 19, 23, 27
3	35	115	8	43, 51, 59, 67, 75, 83, 91, 99
4	115	403	16	131, 147, $\dots$, 371

Table 1: The first levels. Block endpoints are half-open; the labels stay away from both endpoints by margins tending to infinity.

Lemma 5.1 (Global separation). *If $s \neq t$ are finite binary words and $m = \max(|s|, |t|)$, then*

$$|p_s - p_t| \geq 2^m. \quad (13)$$

Proof. If both words have length m , their distinct binary values and formula (10) give the bound. If the levels differ, the label at the higher level is to the right of every label at a lower level. For $m \geq 1$, even its distance from the last label of the immediately preceding level is

$$\min T_m - \max T_{m-1} = 2^{m+1}.$$

All earlier labels are still farther away. This is stronger than the claimed bound. $\square$

5.2 Branches, positivity, and translated intersections

For $x \in 2^\omega$, write $x \upharpoonright n$ for its prefix of length n and set

$$A_x = \{p_{x \upharpoonright n} : n \geq 0\}. \quad (14)$$

Every branch has exactly one point in every B_n . Therefore

$$\sigma(A_x) = \omega \in \mathcal{U}, \quad A_x \notin \mathcal{I}_{\mathcal{U}}. \quad (15)$$

The identity shift alone witnesses positivity.

Proposition 5.2 (Quantitative translated intersections). *Let $x, y \in 2^\omega$. If $k \neq 0$, define*

$$N(k) = \lfloor \log_2 |k| \rfloor + 1.$$

Then, without any requirement that $x \neq y$,

$$A_x \cap \tau_k(A_y) \subseteq [0, b_{N(k)}), \quad |A_x \cap \tau_k(A_y)| \leq N(k). \quad (16)$$

If $x \neq y$ and $d = \min\{j : x(j) \neq y(j)\}$, then

$$|A_x \cap A_y| = d + 1. \quad (17)$$

Proof. For $k \neq 0$, an intersection point gives $p_{x \upharpoonright n} = p_{y \upharpoonright r} + k$. The two node labels are distinct. By Lemma 5.1,

$$|k| \geq 2^{\max(n, r)}.$$

Since $2^{N(k)} > |k|$, both n and r are smaller than $N(k)$. Thus the intersection is contained in the first $N(k)$ levels, all of which lie below $b_{N(k)}$. There is at most one point of A_x in each such level, proving (16).

For $k = 0$, equality of node labels forces equality of the words, since distinct levels are in disjoint blocks and labels within a level are distinct. Two different infinite branches have the same prefixes exactly at lengths $0, 1, \dots, d$. This includes the empty prefix and explains the $+1$ in (17). $\square$

The cutoff is a uniform bound for each fixed nonzero shift, independent of both branches. It is not asserted to be optimal. The dependence on k is essential, and the zero shift is treated separately because two branches can share an arbitrarily long finite prefix.

Proposition 5.3 (The perfect family). *The map $\Phi(x) = A_x$ is a continuous injection from 2^ω into 2^ω . Its image is compact and perfect and has cardinality $\mathfrak{c}$.*

Proof. For a coordinate q that is not a node label, membership $q \in A_x$ is constantly false. If $q = p_s$, it is equivalent to s being a prefix of x , a clopen condition on x . Hence every coordinate map is continuous, so Φ is continuous in the product topology. Distinct branches differ at some prefix, and their sets consequently differ at the corresponding label; thus Φ is injective.

A continuous injection from the compact space 2^ω into the Hausdorff space 2^ω is a homeomorphism onto its image. The image is therefore compact, has no isolated points, and has the cardinality of Cantor space. $\square$

Proof of Theorem 1.2. Use the partition (9) in (8). Proposition 4.3 establishes all assertions about the ideal, including tallness and the category of its positive restrictions. Equation (15) establishes positivity of every branch. Proposition 5.2 gives the finite-intersection properties, and Proposition 5.3 gives the required perfect parametrization. Since $\text{Fin} \subseteq \mathcal{I}_{\mathcal{U}}$, the family is in particular $\mathcal{I}_{\mathcal{U}}$ -TAD. This proves the stated affirmative answer to Problem 1.1. $\square$

6 The ultrafilter trace and the quotient Boolean algebra

The increasing margins in (11) have another purpose: a fixed translation eventually keeps every selected point in its own block. This identifies the ideal on every branch exactly.

For $S \subseteq \omega$, define

$$A_x[S] = \{p_{x \upharpoonright n} : n \in S\}.$$

Proposition 6.1 (Exact trace formula). *For every $x \in 2^\omega$, $S \subseteq \omega$, and $k \in \mathbb{Z}$,*

$$\sigma(\tau_k(A_x[S])) = {}^* S. \quad (18)$$

Consequently,

$$A_x[S] \in \mathcal{I}_{\mathcal{U}} \iff S \notin \mathcal{U}. \quad (19)$$

Proof. Whenever $2^n > |k|$, the margin estimates show that $p_{x \upharpoonright n} + k$ is still in B_n . Thus all sufficiently late selected points retain their block index. The finitely many early selected points contribute only finitely many block indices before and after translation, proving (18).

If $S \notin \mathcal{U}$, every support in (18) is outside $\mathcal{U}$, so $A_x[S] \in \mathcal{I}_{\mathcal{U}}$. If $S \in \mathcal{U}$, the identity shift has support exactly S , and hence $A_x[S] \notin \mathcal{I}_{\mathcal{U}}$. $\square$

Thus $n \mapsto p_{x \upharpoonright n}$ transports the dual ideal of $\mathcal{U}$ to $\mathcal{I}_{\mathcal{U}} \upharpoonright A_x$. In particular, this trace is a maximal ideal on A_x . Notice that this does not make the whole ideal maximal: there are many different positive branches.

Recall that $\mathcal{P}(\omega)/\mathcal{I}_{\mathcal{U}}$ identifies A and D when $A \triangle D \in \mathcal{I}_{\mathcal{U}}$. An *atom* in a Boolean algebra is a nonzero element with no strictly intermediate element between it and zero.

Corollary 6.2. *Each $[A_x]$ is an atom in $\mathcal{P}(\omega)/\mathcal{I}_{\mathcal{U}}$. The elements $[A_x]$ are distinct and pairwise disjoint, so the quotient has at least $\mathfrak{c}$ atoms. Distinct free ultrafilters give distinct ideals in the construction.*

Proof. Every subset of A_x has the form $A_x[S]$. Exactly one of S and its complement belongs to $\mathcal{U}$. By (19), either $A_x[S]$ is zero in the quotient, or its complement relative to A_x is zero there. This proves atomicity. More generally, an element below $[A_x]$ can be represented by a subset of A_x , by intersecting a representative with A_x .

For $x \neq y$, the intersection $A_x \cap A_y$ is finite, so the two nonzero atoms are disjoint and distinct. Finally, if $\mathcal{U} \neq \mathcal{V}$, choose S belonging to one ultrafilter and not the other. Formula (19), using any one fixed branch, distinguishes $\mathcal{I}_{\mathcal{U}}$ from $\mathcal{I}_{\mathcal{V}}$. $\square$

The occurrence of continuum many atoms is compatible with the countability of the ground set. These atoms live in a quotient by an ideal; the representing sets are not literally disjoint, only disjoint modulo the ideal. No claim is made that these are all the atoms.

7 Abstract densities with this exact null ideal

7.1 The meaning of abstract upper density

An *abstract upper density* is a map $\delta : \mathcal{P}(\omega) \rightarrow [0, 1]$ such that $\delta(\omega) = 1$, finite sets have value zero, δ is monotone, and

$$\delta(A \cup D) \leq \delta(A) + \delta(D).$$

It is *rich* when its range is the whole interval $[0, 1]$, and translation invariant when $\delta(\tau_k(A)) = \delta(A)$ for every k . This is the framework of [2, 1]. It does not require agreement with ordinary asymptotic density, finite additivity, or a dilation identity. The distinction matters for our example.

7.2 A maxitive realization and prescribed branch values

The next argument is an application of the established maximal-TAD construction [1, Theorem 3.1], related to the maximal-family method of [2]. We give the formula and verify it, rather than claiming the general mechanism as a new theorem.

Theorem 7.1. *Let $w : 2^\omega \rightarrow (0, 1]$ be any surjection. There is a translation-invariant rich abstract upper density δ with*

$$\{A : \delta(A) = 0\} = \mathcal{I}_{\mathcal{U}}, \quad \delta(A_x) = w(x) \quad (x \in 2^\omega). \quad (20)$$

It can be chosen maxitive:

$$\delta(A \cup D) = \max\{\delta(A), \delta(D)\}. \quad (21)$$

Proof. Extend $\{A_x : x \in 2^\omega\}$ to a maximal $\mathcal{I}_{\mathcal{U}}$ -TAD family $\mathcal{M} \subseteq \mathcal{I}_{\mathcal{U}}^+$. Zorn's lemma applies because the union of a chain of extending TAD families is again such a family: any two of its members already lie together in one element of the chain.

Define a weight $a : \mathcal{M} \rightarrow (0, 1]$ by $a(A_x) = w(x)$ and $a(C) = 1$ on any remaining members. All weights are strictly positive. For $A \subseteq \omega$, let

$$W(A) = \{C \in \mathcal{M} : \text{there is } k \in \mathbb{Z} \text{ with } C \cap \tau_k(A) \notin \mathcal{I}_{\mathcal{U}}\},$$

and set

$$\delta(A) = \sup\{a(C) : C \in W(A)\}, \quad \sup \emptyset = 0. \quad (22)$$

We verify each required property.

If $A \subseteq D$, then $W(A) \subseteq W(D)$, so δ is monotone. A finite A has no witnesses, so its value is zero. Every member of $\mathcal{M}$ witnesses ω , and w takes the value 1; hence $\delta(\omega) = 1$.

For any fixed C and k , the intersection with a shifted union is

$$C \cap \tau_k(A \cup D) = (C \cap \tau_k(A)) \cup (C \cap \tau_k(D)).$$

A union belongs to an ideal exactly when both constituent sets do. It follows that $W(A \cup D) = W(A) \cup W(D)$. Taking suprema proves (21), which implies subadditivity.

For translation invariance, $\tau_k(\tau_h(A))$ differs finitely from $\tau_{k+h}(A)$, and adding or removing finitely many points does not change whether an intersection is positive. As k ranges over $\mathbb{Z}$, so does $k + h$. Thus $W(\tau_h(A)) = W(A)$.

If $A \in \mathcal{I}_{\mathcal{U}}$, every shift and every intersection of a shift with a member of $\mathcal{M}$ belongs to $\mathcal{I}_{\mathcal{U}}$, so $W(A) = \emptyset$. Conversely, suppose $A \notin \mathcal{I}_{\mathcal{U}}$ and $W(A) = \emptyset$. Then A is translated almost disjoint from every member of $\mathcal{M}$. The condition is symmetric: shifting $C \cap \tau_k(A)$ back by $-k$ gives $A \cap \tau_{-k}(C)$ modulo a finite set. Also $A \notin \mathcal{M}$, since otherwise it witnesses itself with $k = 0$. Therefore adjoining A would strictly enlarge $\mathcal{M}$, a contradiction. Hence every positive A has a witness, and its density is strictly positive. This proves the exact-null-ideal assertion.

Finally, $W(A_x) = \{A_x\}$. The inclusion of A_x follows from positivity; no other member of the TAD family can witness it. Therefore $\delta(A_x) = a(A_x) = w(x)$. Surjectivity of w supplies every value in $(0, 1]$, and the empty set supplies zero. $\square$

A surjection w can be chosen very concretely: take $r(x) = \sum_{j \geq 0} x(j)2^{-j-1}$, and replace its value 0 by 1. This has range $(0, 1]$. Neither continuity nor measurability of the weights is required.

7.3 Why finite additivity is impossible

Proposition 7.2. *There is no finitely additive probability measure on all subsets of ω whose null ideal is exactly $\mathcal{I}_{\mathcal{U}}$.*

Proof. Suppose μ were such a measure. Every A_x would have positive measure, and all intersections $A_x \cap A_y$ with $x \neq y$ would have measure zero. For any finite list of distinct branches, successively remove the intersections with earlier branches. This makes the list literally disjoint without changing any of its measures. Therefore

$$\sum_{j=1}^r \mu(A_{x_j}) \leq 1.$$

For each positive integer m , at most m branches can have measure at least $1/m$. But every positive real number is at least $1/m$ for some m . Thus only countably many branches could have positive measure, contradicting their continuum cardinality. $\square$

This is the usual countable-chain-condition obstruction for a measure algebra, applied directly. In particular, the nonadditivity of the maxitive density is not an accidental weakness of the formula. Moreover, any function with exact zero set $\mathcal{I}_{\mathcal{U}}$ cannot be Baire measurable: a Baire-measurable real-valued function has a zero set with the Baire property, since that zero set is a countable intersection of preimages of open intervals.

8 Comparison with ordinary density

For clarity, define upper Banach density here by

$$d^*(E) = \limsup_{L \rightarrow \infty} \sup_{a \in \omega} \frac{|E \cap [a, a + L]|}{L}, \quad L \in \mathbb{N}_{>0}.$$

We need only its defining estimates, not any structural theorem about this density. Let $\mathcal{Z}_B = \{E : d^*(E) = 0\}$.

Proposition 8.1. *The ideals $\mathcal{I}_\mathcal{U}$ and $\mathcal{Z}_B$ are incomparable under inclusion. Every branch A_x has upper Banach density zero but is $\mathcal{I}_\mathcal{U}$ -positive. Conversely, $\mathcal{I}_\mathcal{U}$ contains a set of upper Banach density one.*

Proof. Let $T = \bigcup_n T_n$ be the set of all tree labels. Its successive gaps tend to infinity by (11) and (12). Given a positive integer M , remove a finite initial set $F \subseteq T$ so that consecutive points of $T \setminus F$ are at least M apart. Any interval of length L contains at most $L/M + 1$ such points, so

$$\sup_a \frac{|T \cap [a, a + L]|}{L} \leq \frac{1}{M} + \frac{|F| + 1}{L}.$$

First let $L \rightarrow \infty$, then $M \rightarrow \infty$. This gives $d^*(T) = 0$ and hence $d^*(A_x) = 0$. Positivity was proved in (15).

For the other direction, the four residue classes modulo 4 partition ω , so exactly one of them, say R_r , belongs to $\mathcal{U}$. Let $S = R_{r+2 \bmod 4}$ and put

$$E = \bigcup_{n \in S} B_n.$$

Fix $k \in \mathbb{Z}$. Since $L_n \rightarrow \infty$, all sufficiently late blocks and their neighbors have length larger than $|k|$. Translating a late block by k can therefore meet only itself and its immediate neighbors. Up to finitely many block indices,

$$\sigma(\tau_k(E)) \subseteq S + \{-1, 0, 1\}.$$

The set on the right misses R_r . Consequently the support has finite intersection with the member R_r of $\mathcal{U}$ and cannot belong to $\mathcal{U}$. This is true for every k , so $E \in \mathcal{I}_\mathcal{U}$.

On the other hand, E contains the full intervals B_n for infinitely many n , with lengths tending to infinity. For each fixed L , some such interval has length at least L ; it contains an interval of length L entirely inside E . Thus the supremum in the definition of d^* is 1 for every L , and $d^*(E) = 1$. $\square$

It follows in particular that the density in Theorem 7.1 does not extend ordinary asymptotic density: it assigns a positive value to sets A_x whose asymptotic density is zero. Nor can it dominate upper Banach density, because it vanishes on the set E just constructed. These observations locate the result inside the *abstract* density framework and prevent an unintended stronger interpretation.

9 Extension to countable permutation actions

The mechanism is not specific to integer addition. Countability of the transformations and finiteness of their preimages of finite sets are the ingredients used in the category proof. The tree can also be built by finite avoidance rather than by the explicit numerical formula.

Theorem 9.1. *Let a countable group G act by permutations on a countably infinite set X . In ZFC there is a proper, tall, G -invariant ideal $\mathcal{I}$ on X without the Baire property and a perfect family $\{A_x : x \in 2^\omega\} \subseteq \mathcal{I}^+$ such that*

$$A_x \cap gA_y \text{ is finite} \quad (x \neq y, g \in G).$$

The restriction of $\mathcal{I}$ to each positive set also lacks the Baire property. If the action is free, $A_x \cap gA_x$ is finite for every $g \neq e$ as well.

Proof. Choose an enumeration of X and increasing finite symmetric sets $G_n \subseteq G$ with $e \in G_n$ and $\bigcup_n G_n = G$. We recursively build finite blocks B_n and distinct labels p_s for all binary words s , putting the labels of words of length n into B_n .

At stage n , the previously constructed blocks form a finite set. Let e_n be the first point of the enumeration not yet used in a block. Reserve it as a filler. Choose the 2^n labels of the new level one at a time. A new label must avoid all previous blocks, the filler e_n , the already chosen labels at this level, and

$$\{gp_t : g \in G_n, t \text{ labels a previously chosen tree node}\}.$$

Only finitely many points are forbidden at each choice, so the infinite set X always supplies another point. Symmetry of G_n ensures that avoiding $p_s = gp_t$ also avoids $p_t = gp_s$. When the level has been chosen, let B_n be its labels together with e_n .

The blocks are pairwise disjoint and finite. They cover X : at every stage the first unused point is placed in a block, so every fixed point of the enumeration is eventually used. For distinct tree nodes s, t , the construction guarantees

$$p_s \neq gp_t \quad \text{whenever } g \in G_N \text{ and } \max(|s|, |t|) \geq N. \quad (23)$$

Indeed, at the later of the two selection times the element g and its inverse belong to the current finite symmetric set, and the relevant point is forbidden. This also covers two distinct nodes chosen at the same level.

Fix a free ultrafilter $\mathcal{U}$ on ω , use block support for this partition, and put

$$\mathcal{I} = \{A \subseteq X : \sigma(gA) \notin \mathcal{U} \text{ for every } g \in G\}.$$

It is proper because the identity transformation sends X to X . The fusion lemma and Corollary 4.2 apply to the permutation maps, giving the category and tallness assertions. For $h \in G$, reindexing g by gh proves $hA \in \mathcal{I}$ if and only if $A \in \mathcal{I}$.

Define $A_x = \{p_{x|n} : n \geq 0\}$. It has exactly one point in each block and is positive. For $g \in G_N$, an equality between a point of A_x and a point of gA_y either involves the same tree node or, by (23), involves two nodes at levels below N . Distinct branches have only finitely many common nodes, and there are only finitely many nodes at levels below N . This proves the required finite intersection. In a free action a nonidentity g fixes no point, so the same-node possibility disappears even for $x = y$.

Finally, membership of a label p_s in A_x is exactly the clopen prefix condition, while a filler point belongs to no branch. The same compactness argument as in Proposition 5.3 gives a perfect family. $\square$

For a nonfree action the final self-intersection assertion need not hold: the identity action fixes every branch. This is why the theorem separates distinct-branch intersections from self-intersections. The main result on ω used explicit truncated shifts; the theorem above concerns genuine permutation actions and does not silently identify the two settings.

10 Verification, reproducibility, and proof audit

10.1 What the program checks

The accompanying `verify.py` uses only the Python standard library and exact integer arithmetic. It contains explicit checks that remain active even under Python's optimization flag. A failed check raises an error and terminates with a nonzero exit status.

The recorded run completed **13,012,145 successful checks** with Python 3.13.5. These include the replay of the random draw, the block boundary formula, uniqueness and placement of all 8,191 tree labels through level 12, their endpoint margins, and adjacent-label separation. For every nonzero integer shift from -512 to 512 , the program scans all those labels for collisions and checks the theoretical cutoff.

All 256 binary branch prefixes of length 8 are also tested. Their zero-shift intersections are checked against the exact common-prefix formula. Every ordered pair, including identical prefixes, is checked for every nonzero shift from -31 to 31 . Both the cardinality bound and the location bound are verified. A further set of tests checks same-block transport under shifts from -256 to 256 whenever the margin hypothesis applies.

Finally, eight finite fusion constructions are run through 24 stages each, using separate reproducible seeds and auxiliary blocks of lengths $3, 4, 5, \dots$. The program checks every finite-stage separation condition, then independently verifies that each even-odd transformed support overlap is confined to the predicted early stages. The auxiliary blocks keep the test inexpensive; the written fusion lemma applies to *every* finite-block partition, including the main exponentially growing one.

10.2 Files and reproducibility

File	Purpose
<code>article.tex</code> , <code>article.pdf</code>	Source and compiled article.
<code>verify.py</code>	Exact finite verification program.
<code>random_selection.json</code>	Original area pool, seed, and selected index.
<code>verification_results.json</code>	Parameters, check counts, and limitations.
<code>verification_report.txt</code>	Readable summary of the recorded run.
<code>tree_levels.csv</code>	Exact block and label data through level 12.
<code>shift_collisions.csv</code>	Collision counts for the finite tree and tested shifts.
<code>fusion_certificates.json</code>	Selected finite-block subsequences in the fusion tests.
<code>literature_search.md</code>	Sources consulted and search limitations.
<code>README.md</code> , <code>Makefile</code>	Build and execution instructions.

To rerun the checks, execute `python3 verify.py`. To rebuild the PDF, execute `latexmk -pdf article.tex`, or run `pdflatex article.tex` twice. No network access is needed for these steps once the archive has been obtained. The compiled article uses ordinary LaTeX packages; its bibliography is embedded in the source.

10.3 The infinitary checks are mathematical, not computational

No finite object is substituted for a free ultrafilter. On a finite set every ultrafilter is principal, which would invalidate the finite-intersection contradiction in the fusion lemma. Accordingly, the program does not claim to query $\mathcal{U}$, test ideal membership, or verify failure of the Baire property.

The main logical dependencies can be audited as follows. First, the category criterion produces a finite-block obstruction from the assumption of the Baire property. Second, the fusion lemma defeats every such obstruction using finite fibers and one fixed free ultrafilter. Third, the orbit-core definition enforces invariance; truncated-shift compositions are used only modulo finite sets. Fourth, the branch sets are positive because they hit every block, not because of an assumption about their density. Fifth, global separation treats collisions between different levels as well as within a level. Finally, continuity and injectivity, rather than a bare cardinality assertion, produce the perfect family.

The proof of the density corollary has two further important points: all weights are strictly positive, so a nonempty witness set cannot have supremum zero; and maximality is invoked for exactness of the null ideal, not merely for the availability of many prescribed values. The group extension uses symmetric finite sets of group elements so that avoidance works in both directions.

11 Conclusion and remaining scope

The construction separates two tasks that initially look in tension. Ultrafilter block supports make every full block selector positive, while a translation core makes the ideal invariant. Neither operation requires selectors to be close together. By leaving enough room inside each block for widely separated binary-tree labels, one obtains continuum many positive selectors with only finite translated intersections. The fusion lemma then supplies the missing category argument for the core.

Thus the mathematical argument presented here gives an affirmative construction for the stated Question 1, and in fact a perfect-family version with explicit bounds. The further consequences show that the example is tall, has maximal ultrafilter traces on its branches, admits a rich maxitive abstract density with its exact null ideal, and has an analogue for countable permutation actions.

There are definite limits. The construction does not settle whether every translation-invariant ideal is the null ideal of a rich invariant abstract density. It does not construct such a density for a translation-maximal ideal. In fact $\mathcal{I}_{\mathcal{U}}$ is not translation maximal: fix distinct branches A_x, A_y and form the invariant ideal generated by $\mathcal{I}_{\mathcal{U}}$ together with all shifts of A_x . If that ideal contained ω , then A_y would be covered by an ideal-small set and finitely many shifts of A_x . Its intersections with those shifts are finite, forcing $A_y \in \mathcal{I}_{\mathcal{U}}$, a contradiction. The generated ideal is therefore proper and strictly larger.

Nor is a definable or computable replacement for the free ultrafilter provided. A proof-assistant formalization and an independent mathematical review remain distinct verification tasks, and the literature-priority question remains separate from the correctness of the explicit argument.

A Random-area selection record

The original decimal seed was

209442498745758881649029240816391254350.

It was generated using `secrets.randbits(128)`. A local instance of `random.Random(seed)` then selected `randrange(24)`, obtaining **22**. The recorded UTC time was September 20, 2026, at 21:02:49.492499 UTC. The complete pool, in its original order, was:

Index	Area
0	Analytic number theory
1	Algebraic number theory
2	Finite group theory
3	Commutative algebra
4	Discrete geometry
5	Extremal graph theory
6	Probability and random processes
7	Dynamical systems
8	Approximation theory and special functions
9	Matrix analysis and inequalities
10	Topology and knot theory
11	Formal languages and automata
12	Order theory and lattice theory
13	Enumerative combinatorics
14	Additive combinatorics
15	Optimization and convex analysis
16	Coding theory
17	Algebraic geometry
18	Functional equations
19	Combinatorial game theory
20	Spectral graph theory
21	Numerical analysis
22	Set theory
23	Mathematical logic and computability

The selected mathematical area was random; the subsequent choice of Problem 1.1 within that area was based on its precise published formulation and the availability of a plausible constructive approach. Random selection does not guarantee that another researcher has not chosen the same problem.

B A compact proof map

For a reader checking the central claim independently, the whole argument can be reduced to the following sequence of assertions.

Start with any free $\mathcal{U}$ and the blocks

$$B_n = \left[\frac{4^n - 1}{3} + 2(2^n - 1), \frac{4^{n+1} - 1}{3} + 2(2^{n+1} - 1) \right) \cap \omega.$$

Let $\sigma(A) = \{n : A \cap B_n \neq \emptyset\}$ and impose $\sigma(\tau_k(A)) \notin \mathcal{U}$ simultaneously for every integer k . The resulting family is an ideal, is proper at the identity shift, and is invariant by composition modulo finite sets.

To test the category assertion, begin with an arbitrary finite-block partition (Q_j) . Select a subsequence so that the first n translated supports of the new block avoid the first n translated supports of all preceding selected blocks. This is possible because a finite union of the B_n is finite and each translation has finite preimages. The even and odd selected unions have finite cross-intersection of their block supports under every pair of fixed shifts. They cannot both have a shifted support in a free ultrafilter. One therefore belongs to the ideal and contains infinitely many Q_j , contradicting the category criterion under a Baire-property assumption.

Finally, put

$$A_x = \left\{ \frac{4^n - 1}{3} + 2(2^n - 1) + 2^n(1 + \text{val}(x \upharpoonright n)) : n \geq 0 \right\}.$$

These sets are positive full selectors. Distinct node labels with maximum level m differ by at least 2^m . A fixed nonzero difference can therefore occur at only finitely many levels; difference zero between distinct branches is confined to their common prefixes. Coordinate membership is a clopen prefix condition, so the family is perfect. These statements are sufficient for the main theorem without using any of the density or group-action extensions.

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